

A Spatiotemporal Compactness Pattern Analysis of Congressional Districts to Assess Partisan Gerrymandering: A Case Study with California and North Carolina

Chao Fan, Wenwen Li, Levi J. Wolf & Soe W. Myint

To cite this article: Chao Fan, Wenwen Li, Levi J. Wolf & Soe W. Myint (2015) A Spatiotemporal Compactness Pattern Analysis of Congressional Districts to Assess Partisan Gerrymandering: A Case Study with California and North Carolina, *Annals of the Association of American Geographers*, 105:4, 736-753, DOI: [10.1080/00045608.2015.1039109](https://doi.org/10.1080/00045608.2015.1039109)

To link to this article: <http://dx.doi.org/10.1080/00045608.2015.1039109>



Published online: 14 Jun 2015.



Submit your article to this journal [↗](#)



Article views: 229



View related articles [↗](#)



View Crossmark data [↗](#)

A Spatiotemporal Compactness Pattern Analysis of Congressional Districts to Assess Partisan Gerrymandering: A Case Study with California and North Carolina

Chao Fan, Wenwen Li, Levi J. Wolf, and Soe W. Myint

*GeoDa Center for Geospatial Analysis and Computation, School of Geographical Sciences and Urban Planning,
Arizona State University*

Compactness of a congressional district is a traditional principle in adjudicating gerrymandering claims in political redistricting. During the last decade, many states have used compactness as an important criterion to constrain the presence of gerrymandering in the redistricting process. In this study, we conducted an array of spatiotemporal analyses aiming to evaluate the changes in compactness between the 112th and 113th Congressional districting plans in California and North Carolina, two states that have been well known for their heavy gerrymandering for years. We employed classic shape-based compactness measures, moment-of-inertia-based measures, and measures of partisan bias to assess the districting plans from multiple angles, including irregularity of district boundaries, spatial dispersion, population-weighted shape dispersion, and partisan symmetry. This new and combined use of spatial measures evidenced remarkable increases on the average compactness scores for California's Congress, suggesting general alleviation of the bipartisan gerrymandering in the previous plan. On the contrary, the partisan gerrymandering in North Carolina intensified in the current map, indicated by the substantial decline in the compactness scores for a majority of the districts. Analysis of partisan bias in the districting plans suggested a very slight bias toward Democrats in California in both districting plans. In North Carolina, the partisan advantage shifted from Democrats to Republicans during redistricting. Comparative analysis between the two families of spatial measures revealed the superiority of the moment of inertia family to the classic shape-based indexes for measuring compactness of congressional districts. *Key Words:* compactness, moment of inertia, pattern analysis, political redistricting, population distribution, space-time analysis.

国会选区的密实性，是政治再区划中，裁定不公选区划分宣称的传统原则。过去十年中，诸多州已使用密实性，作为在再区划的过程中，抑制不公选区划分出现的重要准则。在本研究中，我们进行一系列的时空分析，目标在于评估加州和北卡罗莱纳州中，第一百一十二届与第一百一十三届国会选区划分计画中，密实性的改变，而这两州皆以多年来的严重不公选区划分而众所周知。我们运用基于形状的经典密实度方法、根据转动惯量的方法，及党派偏见方法，从多重视角评估选区划分计画，包括分区边界的不规则性、空间散布、人口加权形状散布，以及党派对称性。此一崭新且结合运用的空间方法，显示加州国会的平均密实度比数的显着增加，指出过去计画中，两党对立的不公选区划分的普遍缓和。相对而言，北卡罗莱纳州的党派不公选区划分，在当下的地图中更为剧烈，以多数选区中的密实性比数的显着降低显示之。在选区划分计画中的党派偏见分析，指出在北卡罗莱纳州之中，两个选区划分计画皆轻微对民主党具有偏见。在北卡罗莱纳州的选区再划分中，党派优势从民主党转向共和党。两个空间方法家族之间的比较分析，揭露出转动惯量家族之予以形状为基础的经典指标，在测量国会选区密实性上的优越性。 *关键词：* 密实性，转动惯量，模式分析，政治再区划，人口分布，时空分析。

La compacidad de un distrito congresal es un principio tradicionalmente tenido en cuenta para resolver reclamos de manipulación electoral [gerrymandering] en rezonificación política. Durante la última década, muchos estados han usado la compacidad como criterio importante para contener la presencia de este tipo de manipulación en el proceso de rezonificación. En este estudio, llevamos a cabo una serie de análisis espaciotemporales orientados a evaluar los cambios en compacidad ocurridos entre los planes 112 y 113 de zonificación Congresal para California y Carolina del Norte, dos estados bien conocidos durante años por su alto grado de manipulación electoral. Al respecto, empleamos medidas clásicas de compacidad basadas en forma, medidas basadas en el momento de inercia y medidas de sesgo partidista para evaluar los planes de zonificación desde diversos ángulos, incluyendo la irregularidad de los límites interdistritales, la dispersión espacial, dispersión de forma ponderada por población, y simetría partidista. Este nuevo uso de medidas espaciales combinadas puso en evidencia notables incrementos en el puntaje de la compacidad promedio para el Congreso de California, lo cual

sugiere un alivio general del *gerrymandering* bipartidista del plan anterior. Por el contrario, esta manipulación bipartidista en Carolina del Norte se incrementó en el mapa actual, según lo indica el descenso sustancial de los puntajes de compacidad para la mayoría de los distritos. El análisis del sesgo partidista en los planes de zonificación sugiere un ligero sesgo hacia los Demócratas en California en ambos planes de zonificación. En Carolina del Norte la ventaja partidista cambió de los Demócratas a los Republicanos durante el rediseño de los distritos. El análisis comparativo entre las dos familias de medidas espaciales reveló la superioridad de la familia del momento de inercia sobre los índices clásicos basados en forma para medir la compacidad de los distritos congresales. *Palabras clave:* compacidad, momento de inercia, análisis de patrones, rezonificación política, distribución de la población, análisis espacio-tiempo.

Political redistricting is the process governing how political districts are redrawn. Typically, in the United States, this process occurs every ten years, as new data from the decennial census become available. It has been acknowledged as an important political activity that influences the future political and social representation of states at all levels (Brunell and Grofman 2005; McKee, Teigen, and Turgeon 2006; McKee 2013). Electoral results are often greatly affected by the location of district lines (Gudgin and Taylor 1976). A common practice in the redistricting process is intentional manipulation of district boundaries into configurations that provide an advantage to one party over another, generally known as *partisan gerrymandering*. First coined in 1812, when the Massachusetts Legislature created a salamander-shaped congressional district to benefit the Democratic-Republican Party interest (Figure 1), the term *gerrymandering* is simply a portmanteau of the word *salamander* and Gerry, the last name of the state governor at that time. Gerrymandering affects the electoral process via either “packing” as many votes from one party as possible into several single districts (Lublin 1999) or “cracking” votes from one party over several other districts where the majority supports the other party (Parker 1985; Engstrom 1986). Either form of boundary manipulation can substantially reduce the efficiency of the electoral process by producing wasted votes. In addition, gerrymandering has long been shown to compromise electoral fairness by suppressing public opinion (Sauer 1918; Hayes and McKee 2009; McKee 2013). By decreasing the possibility that the incumbents might lose, gerrymandering can affect the ways in which incumbents behave in legislatures, possibly worsening outcomes for direct representation (Silverberg 1995; McConnell 2000; McKee 2008).

Compactness, which measures the degree to which all elements on a shape are spread out from its center, has long been considered an important criterion in assessing gerrymandering claims (Fryer and Holden 2007; Humphreys 2011). Maximizing compactness of

political districts ensures that the sum of distances from any location to the center of a district is minimized. This prevents the shape of a district from being odd or elongated. It is acknowledged by the Supreme Court that compactness is a “traditional” districting principle that can be used to assess the constitutionality of districting plans (Niemi et al. 1990; Fryer and Holden 2007; Arrow 2012). Many states have used compactness measures, albeit formed in different ways, as a major standard to limit the possibility of gerrymandering. In addition, compactness criteria have been widely employed in a large number of studies in the political redistricting literature (Hess et al. 1965; Garfinkel and Nemhauser 1970; Hojati 1996; Mehrotra, Johnson, and Nemhauser 1998; Bozkaya, Erkut, and Laporte 2003; Bação, Lobo, and Painho 2005; Fryer and Holden 2007; Chambers and Miller 2010; Hodge, Marshall, and Patterson 2010; Angel and Parent 2011; Chou et al. 2012).

Extensive research shows that compactness measures, developed in various forms, can be applied to assess gerrymandering across many successive drawings of district maps (Weaver and Hess 1963; Young 1988; Chambers and Miller 2010; Hodge, Marshall, and Patterson 2010; Li, Goodchild, and Church 2013). One study evaluated and compared the compactness of three different districting plans in California using three compactness measures—circumscribing circle measure, perimeter circle measure, and population polygon measure—each of which represents a different aspect of compactness (Hofeller and Grofman 1990). All three measures suggested that the Masters’ plan (used for the 94th–97th Congress) is more compact than Burton I (used for the 98th Congress) and Burton II (used for the 99th and 100th Congress). Another study examined the congressional districts in ten western states from 1992 to 2012 and tracked the changes in their compactness scores based on the Polsby–Popper test (Miller and Grofman 2013). They used compactness as one of the criteria to assess whether commissions are likely to produce more compact and



Figure 1. A salamander-shaped congressional district drawn by Massachusetts Legislature in 1812 to benefit Democratic-Republican Party. Source: By Elkanah Tisdale (1771–1835), originally published in the *Boston Centinel* (1812); public domain, via Wikimedia Commons.

competitive districts than state legislatures and the courts. As they found out, there was little evidence supporting any consistent relationships between the redistricting authority and the compactness of districts. More recently, the *Washington Post* posted a blog examining the spatiotemporal patterns of political gerrymandering in the past sixty years by looking at how the compactness scores have changed for some of the most gerrymandered districts in the United States. They found that through sixty years, states' districting plans have become less compact over time.

Although there are several attempts in the literature focusing on how the congressional districts evolve through time, many simply examined the mean compactness in the state as a whole. A comparative study examining the compactness changes in individual districts is generally lacking. Quantitative analysis of spatial changes in individual districts over time is difficult

because identifying successor districts across redistricting decades is complicated. In addition, most studies only used a single compactness measure. Doing this, these studies potentially miss the various potential dimensions inherent in compactness as a theoretical construct. Indeed, in the congressional districting context, the concept of compactness involves a variety of distinct concepts that all relate to normative beliefs about what properties congressional district shapes should have, such as boundary regularity, low shape dispersion, and some equitable division of the underlying population distribution. Precise definitions of these normative constructs have been thoroughly discussed and continue to be controversial (e.g., Forest 2013; Quinton 2013; Webster 2013a, 2013b). There is no consensus that a single standard measure of compactness that takes every aspect of compactness into account can exist. To account for this, multiple distinct measures of compactness should be employed

simultaneously for a thorough and accurate assessment of compactness (Niemi et al. 1990). Hence, we conduct a spatiotemporal analysis that aims to assess the changes in districting plans of California and North Carolina—two states that are well known for their bizarrely shaped districts—using multiple compactness measures. Our goal is to evaluate potential partisan gerrymandering in the past and current district plans in the two states as measured by their compactness, and investigate whether and to what extent the gerrymandering has changed over time.

We employed two families of compactness measures—classic shape-based measures and moment-of-inertia (MI)-based measures to evaluate the decadal changes in the districting plans. The well-known shape-based indexes used in this study are the Polsby–Popper and Reock indexes, which measure the geometrical property of a district in terms of how far it is from a circle (or a square for raster models). The MI family, on the other hand, defines compactness as the dispersion of a collection of features around a central point. This family includes normalized moment of inertia (NMI) and its extended version, normalized mass moment of inertia (NMMI) indexes. Together, these two families of compactness measures give a comprehensive assessment of a district's compactness across its many facets, such as boundary irregularity, spatial dispersion, and population-weighted shape dispersion.

Study Area

Figure 2 shows California's congressional districts in the 112th and 113th Congresses colored by party representative. California has historically had frequent and controversial lawsuits over gerrymandering and has been experiencing drastic partisan battles for years (Niemi et al. 1990; Polsby and Popper 1991). In the 112th Congress, the 23rd District, also known as the “Ribbon of Shame,” was often considered one of the narrowest and most irregularly shaped districts in the nation due to its 200-mile winding path along the Pacific coastline (Batkins 2008; Kogan and McGhee 2012; see Figure 2A). As a state commonly considered firmly in the Democratic base, it has witnessed extensive geographic variation in partisan behavior since the 1970s. Traditionally, the state had a north–south divide, with the north supporting Democrats and the south supporting Republicans. This trend has been gradually replaced, however, with an emerging east–west divide, largely due to the population shifts that shaped California's political map dramatically (Douzet, Kousser, and Miller 2008). There are fifty-three congressional districts in total with an average population of 638,430, according to statistics from the 2000 Census. The average population increased to 702,905 in 2010, a significant jump as a result of the rapid population explosion.

In the 112th Congress, although Democratic districts accounted for much less of the state's area—

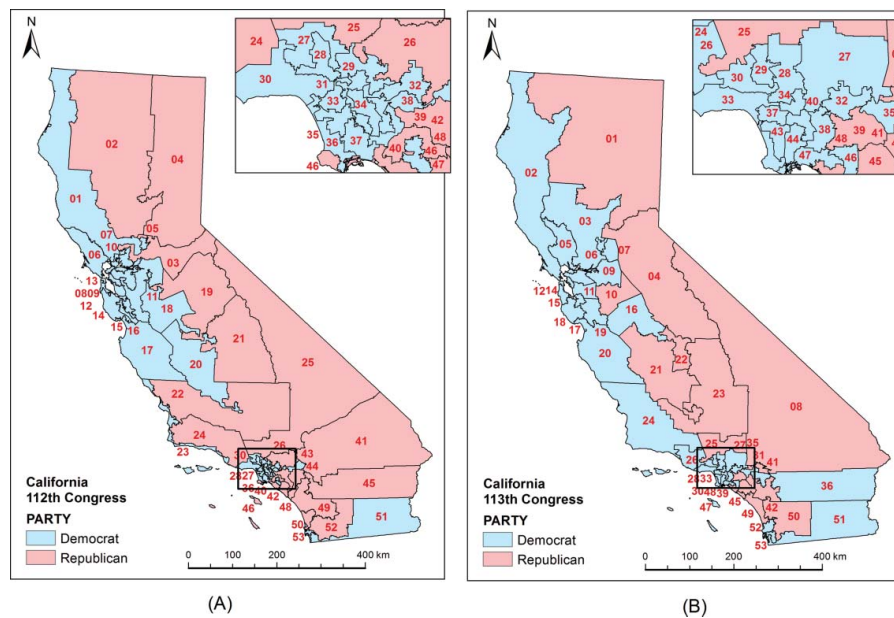


Figure 2. California's (A) 112th and (B) 113th districting plans. Districts in blue are Democratic-held districts and districts in red are Republican-held districts. (Color figure available online.)

about 279,768 km² fewer than Republican districts, they held thirty-four of fifty-three congressional seats with small but strong districts, constituting a solid majority of the state congressional delegation. Growth in Democratic electoral success was apparent in the 113th Congress, with thirty-eight out of fifty-three districts claimed by Democratic representatives, four seats more than under the previous plan. Despite the fact that Republicans are still solid in the eastern part of the state, the area in their hands shrank about 45,457 km² compared to the 112th districting plan.

Figure 3 shows the 112th and 113th districting plans in North Carolina by partisan support. Democrats took seven seats (one more than Republicans) in elections to North Carolina’s 112th Congressional delegation, but their success diminished in the 113th, where they held only four seats (Figure 3). In general, the 113th districting plan appears visually to be less compact than the 112th, which is likely the product of gerrymandering that reduced Democratic success across North Carolina. In addition to concerns about partisan

gerrymandering in North Carolina, significant concerns have manifested about racial gerrymandering, retrogression, and minority vote suppression that are well beyond the scope of this article but are worthy of study in their own right. District 12 in the 112th Congress follows a highway and it is stretched so thin that at some sections it followed I-85 (the interstate highway) nearly perfectly (Figure 3A). This district becomes even narrower and elongated in the 113th plan. Overall, these changes illustrate the need for a multifaceted analysis of the political and spatial dimensions of gerrymandering in these two states.

Methods

Table 1 gives the description, equation, and interpretation of the compactness measures used in this study. They include two classic shape-based measures (i.e., Polsby–Popper and Reock) that have been widely used in both the geography and political science literature. In

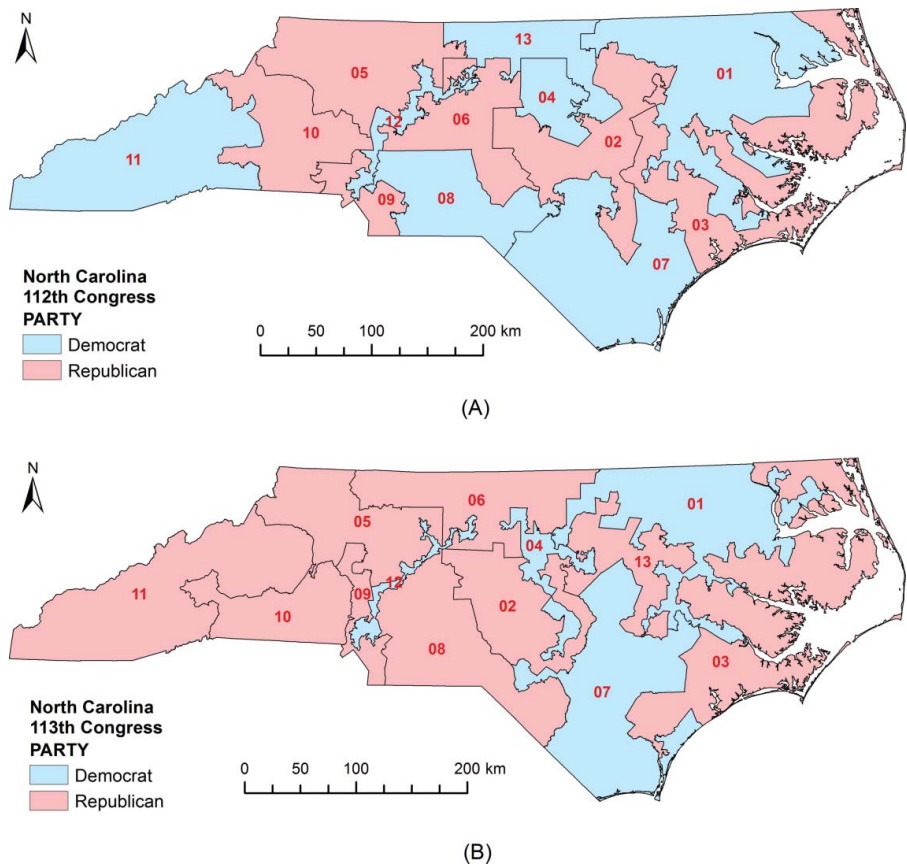


Figure 3. North Carolina’s (A) 112th and (B) 113th districting plans. Districts in blue are Democratic-held districts and districts in red are Republican-held districts. (Color figure available online.)

Table 1. Descriptions of the compactness measures used in this study

Compactness measure	Description	Equation	Variable explanation
Polsby-Popper	The ratio of the area of a shape to the area of a circle that has the same perimeter as the shape	$C_{PP} = \frac{4\pi A}{P^2}$	A is the area of the shape P is the perimeter of the shape
Reock	The ratio of the area of a shape to the area of the minimum enclosing circle of the shape	$C_R = \frac{A}{A_{MEC}}$	A is the area of the shape A_{MEC} is the area of the minimum enclosing circle of the shape
Normalized moment of inertia	The ratio of the area moment of inertia of the circle with the same area as the shape to the moment of inertia of the shape about its centroid	$C_{NMI} = \frac{I_0}{I_i^{G_i}} = \frac{A^2}{2\pi I_i^{G_i}}$	$I_0 = \frac{A^2}{2\pi}$ is the area moment of inertia of the circle with the same area as the shape $I_i^{G_i}$ is the area moment of inertia of the shape about its centroid
Normalized mass moment of inertia	The ratio of the mass moment of inertia of the reference circle with the same mass and effective area as the shape to the mass moment of inertia of the shape	$C_{NMMI} = \frac{I_0'}{I_{i_mass}^{G_i}} = \frac{mA}{2\pi I_{i_mass}^{G_i}}$	I_0' is the mass moment of inertia of the reference circle with the same mass and effective area as the shape $I_{i_mass}^{G_i}$ is the mass moment of inertia of the shape

addition, we used two indexes from the MI family, NMI and NMMI, to quantify the compactness of districts based on the dispersion of (population-weighted) elements from district centers. To measure the potential partisan bias of these new political geographies resulting from redistricting, we estimate partisan bias scores of the Gelman–King model (Gelman and King 1994) using the *JudgIt* package in R (Gelman, King, and Thomas 2010). This provides an overall view of partisan bias in the electoral system that has been applied consistently (e.g., McKee, Teigen, and Turgeon 2006) and provides a link between the selected spatial measures and the political implications of noncompact districts.

Polsby–Popper Index

The Polsby–Popper measure is a compactness measure developed from well-established mathematical insights about the relationship between a shape's area and perimeter, called the isoperimetric inequality or isoperimetric quotient (IPQ). Polsby and Popper (1991) adapted results from the IPQ into an index of compactness to adjudicate gerrymandering claims in political redistricting problems. It corrects previous tests by normalizing the range of the measure to 0 to 1. The Polsby–Popper test, or C_{PP} , is defined as the ratio of the area of a shape to the area of the circle—the reference shape—that has the same perimeter as the shape. Generally, the range of C_{PP} is (0, 1), with higher C_{PP} scores indicating a more compact shape. The most compact shape with this test is a circle,

which receives a compactness score of 1. C_{PP} is one of the most used compactness measures in the political literature because it is simple, invariant to scale, and can be applied to both vector and raster data sets. Because the value of C_{PP} depends to a great extent on the perimeter of the shape being examined, however, it is very sensitive to changes in the boundary of the shape. Thus, C_{PP} is useful in identifying districts with irregular and ragged boundaries.

Reock Index

The Reock index (Reock 1961) is another compactness measure that has had widespread use in the political literature assessing the compactness of districting plans. It is defined as the ratio of a shape's area to the area of the smallest circle containing the shape. Similar to the Polsby–Popper test, the Reock test (or C_R) ranges from 0 to 1, with higher scores indicating a greater similarity to a circle. Despite the discussions in political literature about the potential drawbacks associated with this measure (Young 1988), the Reock index remains a commonly used indicator of district compactness because it is easy to calculate and it allows simple threshold-based analysis to assist policymakers aiming to identify gerrymandering.

Normalized Moment of Inertia

MI methods have been used in shape analysis and districting problems for many years (Hess et al. 1965).

It was widely known as a basic concept in physics and is defined as the second moment of a shape about an axis perpendicular to the shape through its centroid. Use of MI measures has often been restricted to individual problem frames because moment measures were not standardized, limiting their widespread usefulness. A new, robust measure of compactness, the NMI, was recently proposed (Li, Goodchild, and Church 2013). The NMI is a ratio between the MI of a shape and the MI of a circle with the same area. This is a spatial standardization of other MI measures that gives the NMI a range between 0 and 1. The most compact shape, a circle, receives a score of 1, as the MI of the circle and the shape being analyzed are identical. Holding area constant, as the shape being analyzed becomes less compact, the MI of the shape increases, but the MI for the circle with equivalent area stays the same, and the NMI ratio decreases. As a compactness measure, C_{NMI} has prominent computational advantages over traditional compactness measures because it allows the on-the-fly calculation of compactness as the area of a shape changes. It is also not as sensitive to scale changes or irregular boundaries as most compactness measures are (Li, Goodchild, and Church 2013). In addition, C_{NMI} is applicable to multipart polygons and shapes with holes that are often difficult for other compactness indexes to handle. Given its computational efficiency and ability to accurately measure the dispersion of elements, NMI was chosen as an important method in our study.

Normalized Mass Moment of Inertia

The NMMI (Li et al. 2014) is an extension of the NMI that considers the underlying distribution of a particular property (e.g., population, income) on a shape in measuring its compactness. The distribution function, in discrete or continuous forms, is an input parameter to the model. The NMMI of a shape is defined as the ratio of the mass moment of inertia (MMI) of the reference circle with the same mass and effective area as the shape to the MMI of the shape. The range of the NMMI is (0, 1), with higher scores indicating that mass is more concentrated over an area. The NMMI considers both how the mass is distributed and how the distribution is spatially configured. Critically, the NMMI reduces to the regular NMI when mass—population in this context—is evenly distributed over the shape.

Because compactness analysis often implicitly involves assumptions or beliefs about the underlying population distribution, NMMI improves significantly on prior analyses by formally measuring the effects of population dispersion. This makes NMMI a sufficient sociospatial indicator of many different aspects of compactness at the same time. One should always be cautious in making gerrymandering claims based solely on the fact that the shape of a district is elongated or oddly shaped—districts can be elongated and not gerrymandered if the population in the district is very concentrated in a subarea of the shape. Likewise, a district that is compact in shape can still be gerrymandered if the district collects pieces from many dense population centers located at the edges of the district. Therefore, we used NMMI in this study, as it considers both the geometry and population distribution in assessing the compactness of districting plans, rather than focusing on a single dimension.

We developed a hierarchical processing flow to create the population distribution input used to calculate the NMMI (Figure 4). Spatial boundaries of congressional districts (CDs) and population data at the census tract (CT) level were collected from the U.S. Census Bureau. There was a great deal of spatial mismatch between the boundaries of CDs and CTs, making the spatial relationships between CTs and CDs less straightforward. We noticed that, to preserve completeness, the spatial extents of CTs often do not conform to physical boundaries for most coastal areas in both states. This further increased the difficulty of generating the population distribution maps.

Figure 4 shows the workflow for obtaining the boundaries of atomic units and the population data. Atomic units are defined as subdivisions of CTs that are split by the boundaries of CDs. The data we acquired include (1) a text file with spatial relations between the CDs and CTs, (2) population data at the census level, and (3) spatial-explicit polygons delineating the boundaries of the CDs. The text file plays a key role in the entire data processing because it enables the separation of census tracts that are completely within a district from those that are divided by multiple districts. For CTs that are split into multiple districts, we employed the proportion principle to determine the population of the atomic units. That is, the population of an atomic unit is proportional to the area it occupies in its parent CT. The resulting maps were subdivisions of the CDs with population assigned by the atomic units. This procedure was repeated for the four districting plans, respectively.

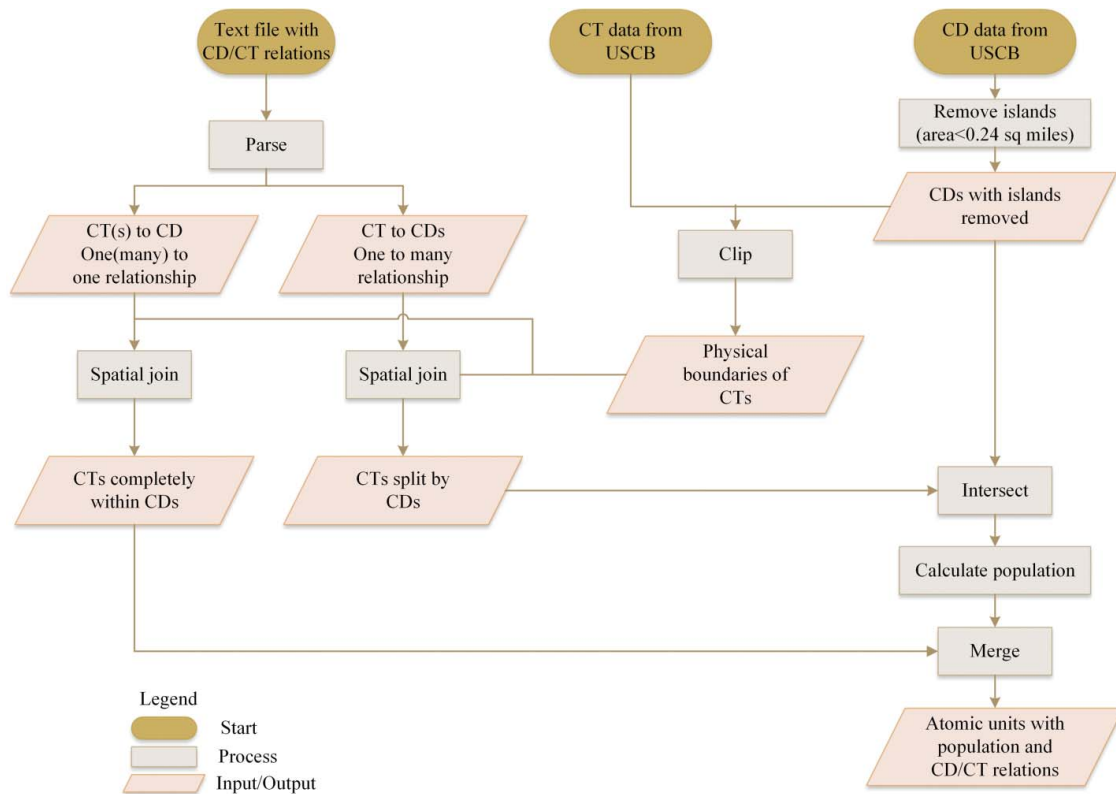


Figure 4. Diagram outlining the main steps for obtaining population and boundaries of atomic units that compose a district. *Note:* CD = congressional district; CT = census tract. (Color figure available online.)

Partisan Bias

In addition to spatial measures, analysis of partisan gerrymandering requires explicit attention to political variables. Historically, interest in developing indicators of partisan gerrymandering focused on two dimensions: bias and responsiveness. Partisan bias in electoral systems has a long history in geographic and political literatures (Brookes 1960; Gudgin and Taylor 1976; Katz 1980). From analysis and attempts to understand the relationship between seats and votes in electoral systems (Taagepera 1973; Tufte 1973) and the later proliferation and refinement of measures of political bias (Grofman 1983; King and Browning 1987; Grofman and Selb 2009), courts in the United States have largely settled on the validity of partisan bias and responsiveness as the two central dimensions of partisan symmetry (Grofman and King 2007). Partisan symmetry occurs when electoral systems provide no unfair advantages to one party over another. Measures of partisan symmetry often establish some relationship between how many seats a party can expect to win in a given election versus a set of

predictor variables such as voter turnout, registration, or candidate incumbency. Using a model of electoral results, bias and responsiveness can be estimated.

Specifically, *JudgeIt*, a political gerrymandering analysis package written in R, is used to analyze the North Carolina and California congressional districting plans. A probability model developed in Gelman and King (1994) relates a vector of district vote share for a party (v), arbitrarily fixed in this situation to Democrats, to some set of predictive factors (X):

$$v = X\beta + \gamma + \varepsilon.$$

This model includes two independent error vectors: one encompassing structural variation due to incomplete knowledge (γ) and another to capture wholly random effects (ε). To estimate β , information about districts was recorded along a few dimensions: Party control of a district, candidate incumbency, whether a candidate was uncontested, voter registration, and turnout of eligible voters are recommended in Gelman and King (1994). This allows us to control for well-understood sources of path dependence and temporal

autocorrelation in the way congresses are divided between parties: Incumbents tend to win, and if a large group of one party's politicians gets elected, it is important to control for their incumbency in analyzing later questions of partisan bias.

Using this model, estimates of β are obtained. Then, holding these estimates constant, we simulate potential elections by varying the information in X while holding β constant. This generates new hypothetical vote outcomes (v^{hyp}), and the seats that Democrats would have won given these hypothetical outcomes are recorded. Because assumptions about the structural variation (γ) are relatively minor, these simulated results are statistically robust and can accurately represent valid or plausible outcomes of the electoral system. Then, the average proportion of seats awarded “over and above what is fair” (Gelman and King 1994, 536) is used to indicate average partisan bias. JudgeIt reports this factor between -1 and 1 , representing a percentage bias toward or against Democrats. For instance, when partisan bias is calculated to be $+0.07$, Democrats win 7 percent more seats than if the electoral system were symmetrical. Because the procedure is fixed to a two-party vote share, this result would also imply that Republicans won 7 percent fewer seats than would be expected with partisan symmetry.

Mapping Districts from the 112th to the 113th Plans

Boundaries of districts changed widely over plans. Therefore, establishing mapping relationships between the two districting plans is critical to examine compactness changes between plans for any individual district. To resolve this, we devised and applied an automated mapping approach using geographic information system techniques. We conducted an intersect analysis on the two redistricting maps from which the overlapped subdivisions were obtained. Each subdivision was a subset of the 112th CDs that was also shared by the 113th districts. Therefore, each subdivision has two Congressional District IDs—one with the 112th and one with the 113th district. If a 112th district has multiple subdivisions after intersecting with district boundaries in the 113th Congress, a one-to-many relationship was established when mapping 112th districts to the 113th, and vice versa. The population for each subdivision was aggregated from the CT data—the same data used earlier. Relationships were created based on the majority population rule. That is, when a one-to-many relation is present, the “to” district is a

113th district that receives the maximal population of all subdivisions in the 112th district. The procedure was performed in a reverse order to handle the cases where districts in the 112th Congress were not divided but aggregated into the 113th. This way, a bidirectional mapping between the 112th and 113th districting maps can be obtained. The final relationships were determined through combining the two relationships.

Results and Discussion

We evaluated the compactness of districting plans in California and North Carolina by calculating four compactness measures. Compactness was evaluated at the levels of the state and individual district as a way to assess the likelihood of a district being gerrymandered. We demonstrated the utility of NMMI in representing the contribution of population distribution in the assessment of district compactness. In addition, we estimated the partisan bias in the congressional districting plans using JudgeIt. The MI family of compactness measures provides a new tool for political districting analysis and offers recourse to new analytical measures for district regularity and representativeness.

Compactness of Districting Plans in California

The consistent increases in the mean scores of four compactness measures suggested that the 113th districting plan is more compact than the 112th plan. The percentage increases in the mean scores were 37.1 percent, 20.6 percent, 17.5 percent, and 17.0 percent for Polsby–Popper, Reock, NMI, and NMMI, respectively. The extrema also increased accordingly, except for the minimum score of NMI. There were barely any discernable changes in the dispersion of the compactness scores, whereas the coefficient of variation (CV), defined as the ratio of the standard deviation to the mean, generally declined due to the growing mean scores. In general, NMI was the highest in magnitude compared to all the other measures, followed by Reock, NMMI, and Polsby–Popper. Note that the values of Polsby–Popper were consistently lower than that of NMI (~ 0.4 unit lower in average scores). This is not surprising given that a large number of districts in California have irregular boundaries. Districts in both plans received much lower NMMI scores than NMI, suggesting that some shape-compact districts in

California were not truly compact when considering the dispersion of population within the district.

We established a spatial relationship between the 112th and 113th districting plans in California following the procedure outlined earlier. More than half of the fifty-three districts in California increased in their compactness scores, and the scores for some districts even doubled in the current plan. This echoed the behavior of descriptive statistics in Table 2: California's current districting plan is more compact than the previous one. Note that the Ribbon of Shame district—District 23 in the 112th plan—had the lowest compactness scores out of all districts. The Ribbon of Shame has no single successor district in the 113th district map, however. The 112th districting plan protected incumbents, with substantial gerrymandering to preserve electoral stability in both parties across the state. As a result, some districts were heavily convoluted to ensure the dominance of one party over the other. It was not until 2008 that the responsibility of drawing CDs was given to an independent Citizens Redistricting Commission, a fourteen-member commission composed of a nearly even number of commissioners from each party. The Independent Citizens Redistricting Commission is a relatively new civic organ designed to move the redistricting process away from state legislatures and state political parties. As the first certified congressional districting by the California Commission, the districts of the 113th Congress in California were more compact. Nevertheless, a recent study suggested very limited evidence that the commissions perform better than legislatures in

producing more compact and competitive districts (Miller and Grofman 2013). Given the fact that only six states across the nation have used commissions to draw their districts and all are relatively new, it still remains an open question whether citizens redistricting commissions will continue to generate substantially more compact or more competitive districts, as the objectives for citizens redistricting commissions vary considerably across states.

Table 3 shows the compactness and JudgeIt bias scores for the 112th and 113th districting plans in California by political party. The compactness measures vary between 0 and 1, as discussed earlier. To aid interpretation, JudgeIt scores are reported as varying between 0 and 1, indicating a percentage bias in favor of one party. Because bias favoring one party presents an equal but opposite bias against the other party, only one bias figure is reported. Between the plans, all of the compactness scores increased for both parties, indicating a general improvement in the overall district compactness across the state. Note that Democrat-held districts tended to be less compact than Republican-held ones in the 112th Congress, as suggested by all of the geometry-based metrics and by NMI. The mean compactness of Democratic and Republican districts grew closer during the redistricting, however. Taking into account the underlying distribution of population through the NMMI, Democratic districts

Table 2. Compactness scores for the 112th and 113th Congressional districts in California

	Min	Max	M	SD	CV
Polsby–Popper					
112th plan	0.024	0.441	0.170	0.089	0.525
113th plan	0.062	0.481	0.233	0.087	0.371
Reock					
112th plan	0.031	0.526	0.306	0.110	0.361
113th plan	0.048	0.624	0.369	0.131	0.356
NMI					
112th plan	0.061	0.857	0.543	0.174	0.321
113th plan	0.053	0.947	0.638	0.165	0.259
NMMI					
112th plan	0.011	0.520	0.224	0.137	0.610
113th plan	0.022	0.592	0.262	0.147	0.561

Note: CV = coefficient of variation; NMI = normalized moment of inertia; NMMI = normalized mass moment of inertia.

Table 3. Compactness and JudgeIt scores for districting plans in California by political party

	M		SD	
	Democrat	Republican	Democrat	Republican
Polsby–Popper				
112th plan	0.142	0.219	0.073	0.094
113th plan	0.231	0.239	0.089	0.078
Reock				
112th plan	0.283	0.347	0.103	0.111
113th plan	0.367	0.373	0.131	0.131
NMI				
112th plan	0.522	0.581	0.167	0.180
113th plan	0.645	0.617	0.163	0.169
NMMI				
112th plan	0.248	0.182	0.149	0.097
113th plan	0.288	0.189	0.151	0.104
JudgeIt				
112th plan	0.074	—	0.023	
113th plan	0.078	—	0.023	

Note: NMI = normalized moment of inertia; NMMI = normalized mass moment of inertia.

showed consistently more population compactness than Republican ones did. We observed a much greater rate of increase—an average of 33 percent—in the compactness for Democrat-dominated districts. Judgment scores demonstrate that there was only a slight bias toward Democrats: They won around 7 percent more seats than expected given their vote share. This remained essentially unchanged after redistricting.

In light of both the spatial and political results, it is evident that the system retained a slight bias in favor of the Democrats even as Democratic districts saw a dramatic improvement in population compactness. It is possible that remaining bias toward Democrats is the result of an unintentional gerrymander (Gudgin and Taylor 1976; Chen and Rodden 2013), which refers to the propensity of electoral districting bodies using politically neutral criteria to generate biased plans (Taylor and Johnston 1979). In California, the relative disparities in the properties of the spatial distribution of population could generate this unintentional partisan bias. Specifically, the Citizens Redistricting Commission had many objectives, like districting compactness, competitiveness, preservation of communities of interest, reuse of existing administrative boundaries, and elimination or reduction in partisan bias. In balancing each of these objectives against their marginal impacts on system-wide partisan bias, it might be true that partisan bias simply could not be reduced without unacceptable decreases in other objectives. That is, they were unable to maximize all relevant dimensions simultaneously and the current map represents the outcome of a complex trade-off between competing demands on districting officials. Overall, the variations in the compactness scores across districts were small (Table 2), indicating that no large spatial variability accompanied these trade-offs during redistricting.

Compactness of Districting Plans in North Carolina

Collectively, the four compactness scores suggested that the 113th CDs in North Carolina were less compact than under the prior districting plan (Table 4). The average compactness scores declined by 14.6 percent, 18.1 percent, 22.1 percent, and 32.2 percent for Polsby–Popper, Reock, NMI, and NMMI, respectively. The extrema resembled the trends of average scores, except for the minimum value of the Polsby–Popper index, which increased slightly. The standard deviations also decreased for all four measures, yet their

Table 4. Compactness scores for the 112th and 113th Congressional districts in North Carolina

	Min	Max	M	SD	CV
Polsby–Popper					
112th plan	0.008	0.293	0.137	0.089	0.646
113th plan	0.009	0.257	0.117	0.082	0.701
Reock					
112th plan	0.116	0.617	0.354	0.115	0.324
113th plan	0.071	0.426	0.290	0.105	0.360
NMI					
112th plan	0.192	0.854	0.594	0.185	0.312
113th plan	0.092	0.714	0.463	0.162	0.351
NMMI					
112th plan	0.051	0.504	0.289	0.143	0.495
113th plan	0.036	0.299	0.196	0.076	0.388

Note: CV = coefficient of variation; NMI = normalized moment of inertia; NMMI = normalized mass moment of inertia.

CVs increased, except NMMI. As with the California plans, the values of the Polsby–Popper index were consistently lower than NMI. The comparatively low scores of the Polsby–Popper index were primarily due to the winding coastal lines and irregular boundaries of some isolated islands in North Carolina.

We employed the same approach to determine the locational mapping between the 112th and 113th Congress in North Carolina. Unlike the case in California, it turned out for North Carolina that no district was dramatically altered in its location. All of North Carolina’s thirteen districts in the 113th Congress have the same IDs as they did in the previous map. The compactness scores decreased significantly for North Carolina between the two plans. This is evidenced by substantial score declines for more than half of the districts across the state. A majority of the newly drawn 113th CDs in North Carolina were less compact than they were in the prior congressional districting plan. An example is District 4 in North Carolina (Figures 5A and 5B), which featured a tremendous decline in compactness scores from the 112th to 113th Congress: 80 percent and 85 percent decreases in Polsby–Popper and NMMI, respectively. Being a heavily Democratic district, District 4 was an excellent example of the packing effect, as what was once a compact district containing two counties now snakes across seven counties, clustering many spatially dispersed Democratic population centers into a single district. Significant gerrymandering was also found around District 12 (Figures 5C and 5D); the “highway” district became even more slender and was hemmed in by six Republican districts in all directions. Some cases

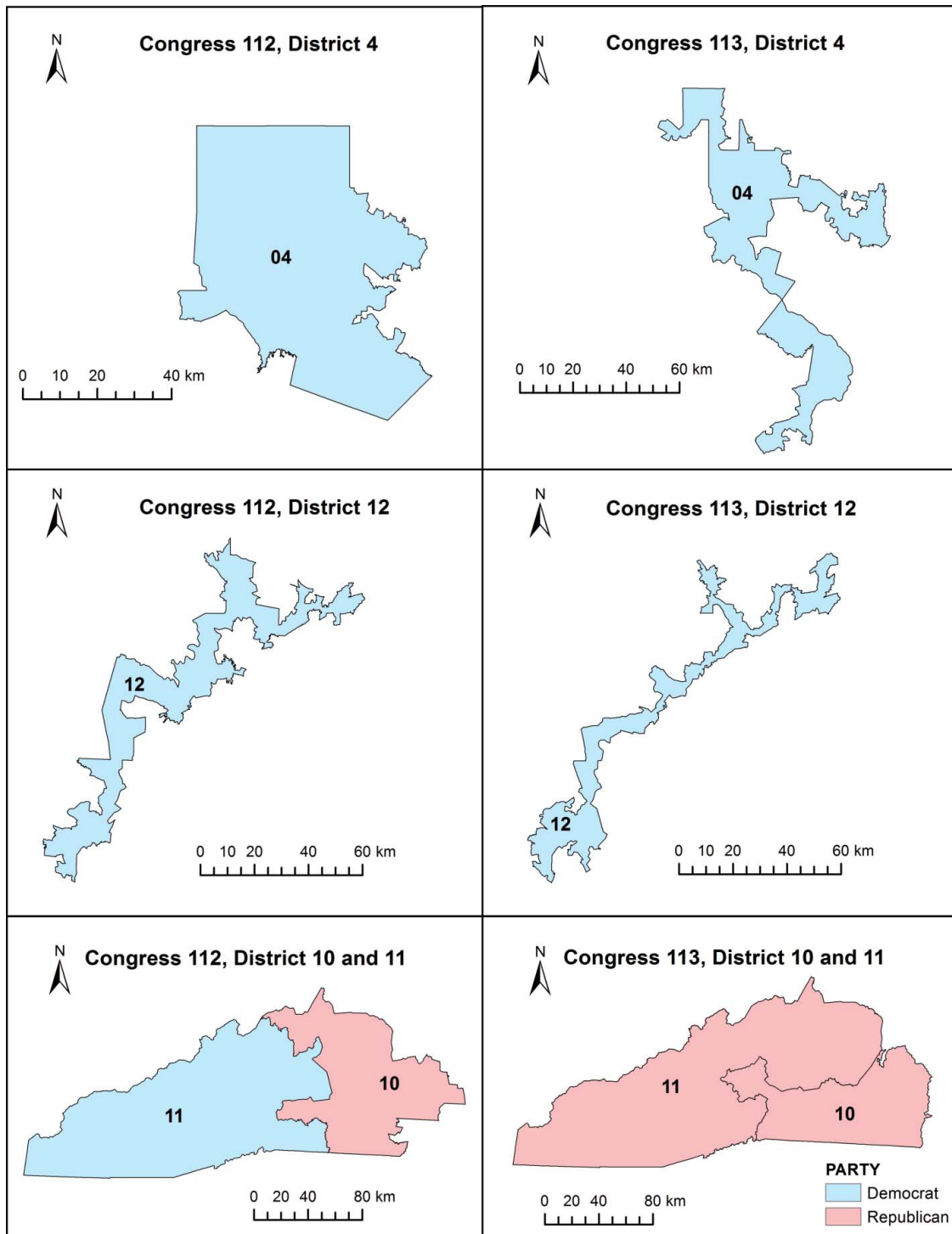


Figure 5. Congressional district maps in North Carolina. Blue and red represent Democrat and Republican, respectively, as in Figure 2. (Color figure available online.)

of cracking, where territorially contiguous communities are split into different districts to dilute representation, were also observed. Part of North Carolina's 11th CD in the previous map was sliced and placed in District 10, a strongly Republican district, causing

Democrats to lose dominance over District 11, which flipped from Democratic to Republican (Figures 5E and 5F).

Table 5 records the compactness and JudgeIt bias scores for North Carolina by political party. Again,

Table 5. Compactness and Judgment scores for districting plans in North Carolina by political party

	M		SD	
	Democrat	Republican	Democrat	Republican
Polsby–Popper				
112th plan	0.148	0.125	0.094	0.080
113th plan	0.051	0.147	0.018	0.082
Reock				
112th plan	0.359	0.349	0.149	0.051
113th plan	0.238	0.314	0.129	0.081
NMI				
112th plan	0.601	0.585	0.215	0.142
113th plan	0.407	0.488	0.231	0.111
NMMI				
112th plan	0.248	0.337	0.160	0.102
113th plan	0.112	0.234	0.064	0.045
Judelt				
112th plan	0.150	—	0.024	
113th plan	—	−0.078	0.040	

Note: NMI = normalized moment of inertia; NMMI = normalized mass moment of inertia.

the compactness indicators vary between 0 and 1 and bias is reported between 0 and 1 as a percentage bias favoring one party. A remarkable decrease in the compactness occurred in Democratic districts—an average 46.6 percent decrease—and there was a much smaller decline (9.9 percent) for Republican districts. The significant reduction in compactness scores likely reflects more pervasive gerrymandering in the 113th Congressional redistricting: Democratic voters were packed into a small number of districts, resulting in oddly shaped or elongated districts. Results of Judgment analysis support this interpretation. In the 112th Congressional elections, Democrats won 15 percent more seats than would be expected given their vote share. By the 113th elections, however, redistricting processes moved this score twenty-three percentage points, turning a sizable systemic advantage into a 7.8 percent disadvantage. Instead, the new districting plan is biased toward Republicans, meaning Republicans won 7.8 percent more seats than expected. As a result, Democratic districts earned only four seats in elections to the 113th Congress and these districts were even less compact than they were in the 112th Congressional map. The NMMI scores for Democratic districts also decreased quite significantly (54.8 percent) from

the 112th Congress, which captures the effects of packing of multiple distant population centers into smaller numbers of Democratic districts. In this case, it is clear that partisan indicators of bias in the electoral system correspond well with the compactness-based analytics of CD gerrymandering tested. In the next section, we further exploit the correlation and discrepancies in the two families of compactness measure in gerrymandering studies.

Comparison of Two Families of Compactness Measure—Geometric and MI—in Gerrymandering Studies

Given the consensus among political geographers that no single standard measure of compactness exists (Manninen 1973; Young 1988) and that multiple measures are needed to represent different aspects of compactness (Niemi et al. 1990), we employed two families of compactness measures—classic shape-based measures and MI measures—to thoroughly evaluate the decadal changes in the compactness in the past and current districting plans. The MI-based metrics permit us to measure the compactness of a district based on the spatial dispersion of (population-weighted) elements from a district's center. It uses two components of a shape—dispersion and population—that are critical for any type of compactness measure (Niemi et al. 1990). The MI family of measures not only provides a more accurate characterization of a district's compactness but also reduces the computational effort, as the two measures used in the MI family have desirable similarities in their mathematical structure. For example, the NMI can be easily upgraded to the NMMI by incorporating a population distribution into the model. Likewise, the NMMI can be easily reduced to the NMI if the population component is not available.

Although there is no clear theory that any correlation should be expected among these measures, we did find notably high correlations between the reduced information version of the NMMI, or NMI, and a standard geometry-based measure, the Reock index. The Pearson's correlation coefficients between NMI and Reock were above 0.85 for the two districting plans in California, and they were much higher (>0.93) for the districting plans in North Carolina. Because the MI family has the ability to accurately represent the geometric aspect of compactness with or without the consideration

of the underlying population distribution, we recommend that the MI-based measures should be widely used as an alternative to the classic shape-based metrics in the compactness pattern analysis.

We identified districts with large differences in the NMMI and Reock to better understand the relationships between the spatial dispersion in the underlying population distribution and how a

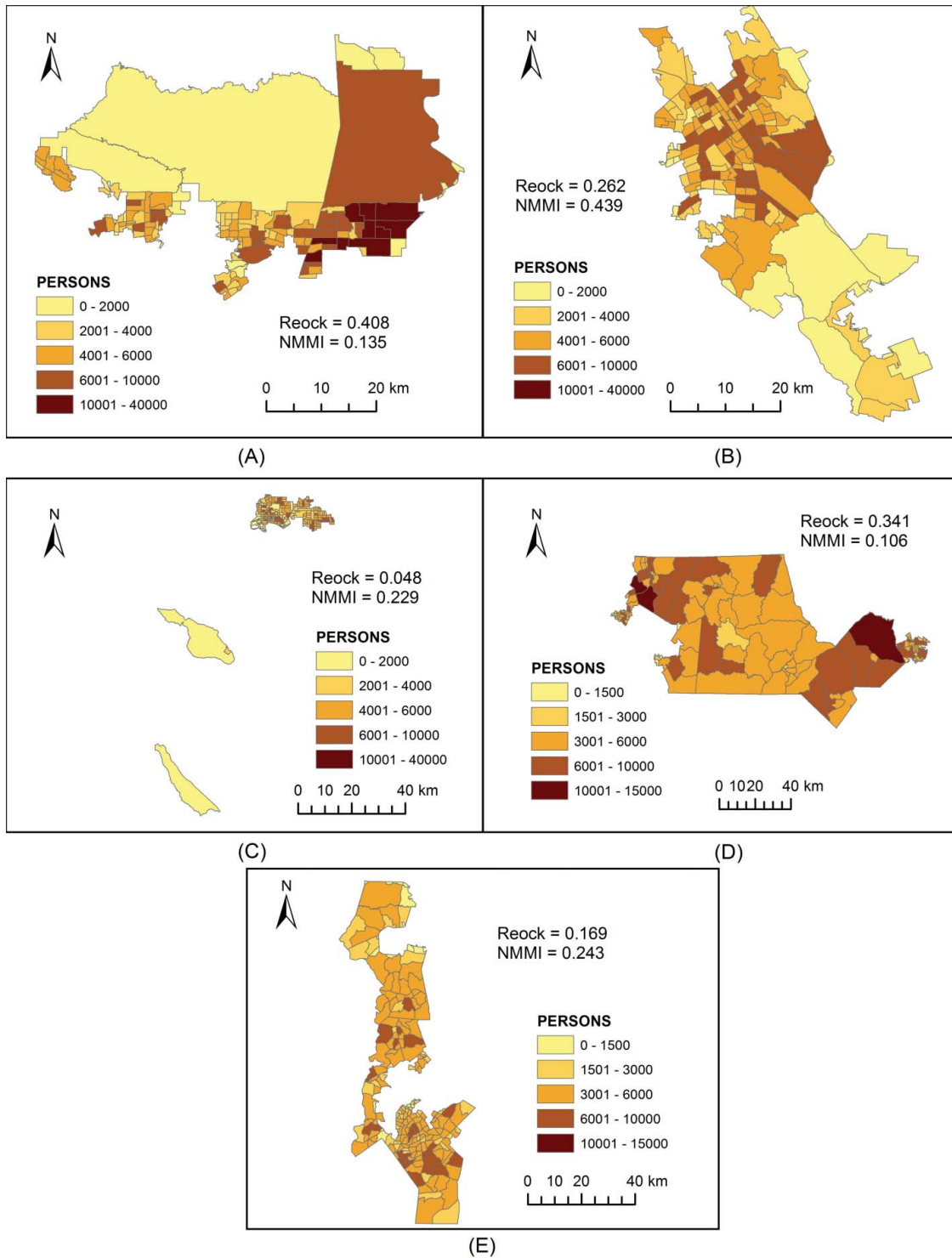


Figure 6. Population distribution maps for congressional districts: (A) District 26 in the 112th Congress in California; (B) District 16 in the 112th Congress in California; (C) District 47 in the 113th Congress in California; (D) District 8 in the 112th Congress in North Carolina; (E) District 9 in the 113th Congress in North Carolina. Note: NMMI = normalized mass moment of inertia. (Color figure available online.)

district's geometric compactness is typically evaluated. While accounting for the difference in the distributions of the two indexes, we observed considerable discrepancies between NMMI and Reock for some districts in both states. One of the major causes for the discrepancies appears to be the existence of natural barriers (e.g., mountains, oceans) in and around the district. For example, for the 26th District in the 112th Congress of California (Figure 6A), the entire northern part of the district is shared by Mount Wilson and Mount Baldy, two natural reserves with very few residences. As a result, population of the district is highly concentrated in the east and a few "antenna" areas in the south of the districts, resulting in a much lower NMMI (0.135) than Reock (0.408). A similar case occurs with District 16 in California's 112th Congress (Figure 6B), where the main mass of population is clustered in the northwest due to the existence of large mountainous areas in the south. Oceans can also be troublesome. Being isolated by the North Pacific Ocean, the 47th District in California's 113th Congress (Figure 6C) has the lowest Reock (0.048) among all of the districts, primarily due to the substantial dispersion of the three islands in the district. The NMMI score, however, turned out to be much higher (0.229), given that the population in this district is mainly clustered in the northern island compared with the other two.

A flip between NMMI and Reock is also seen in North Carolina. For instance, its 8th District in the 112th Congress (Figure 6D) has a moderately compact shape, yielding a Reock of 0.341. The NMMI score, however, dropped to 0.106 simply because the two population centers are located at the very western and eastern ends of the district. The value of NMMI was much higher than Reock for the 9th District in the 113th Congress (Figure 6E). Although elongated in shape, this district has a relatively high NMMI score because its population centers are concentrated in the middle of the district.

These occasional cases where there are stark differences between the NMMI and Reock are important to note. The differences between NMMI and Reock manifest in districts where the compactness of the district shape does not track the underlying population distribution well. Without any significant assumptions about the underlying population distribution, the NMMI incorporates both shape information and the underlying population distribution in characterizing a

district's spatial properties. Through the differences between NMMI and Reock, we can see how shape-based measures might fail to discover cases where a district, although compact in shape, is drawn to pull together disparate populations. Alternatively, the NMMI provides a much better indicator of the spatial relationship between a district's shape and the underlying population distribution in cases where a non-compact district matches the population distribution well. These differences illustrate NMMI's ability to accurately characterize districts in cases that have confounded or resisted analysis using pure shape-based compactness indicators.

Conclusions

Compactness is an important and effective measure that can be used to assess the impartiality of districting plans and adjudicate gerrymandering claims (Fryer and Holden 2007). This article expands previous research regarding the role of compactness in political redistricting but also investigates the progress of political redistricting during the last decade within a spatiotemporal framework. Our results show an increase in the overall compactness level between the 112th and 113th Congress in California, whereas compactness scores declined when comparing the past and current CDs in North Carolina. We also analyzed the compactness changes of political districts by looking at the roughness of boundaries (using Polsby-Popper), the shape compactness (using Reock and NMI), and a factor that reflects the population distribution and the shape compactness (using NMMI). Then, partisan bias was used to link spatial measures to observed or expected electoral outcomes. We believe that this multiangle examination of CDs gives us a comprehensive view about the spatial implications of partisan gerrymandering.

Specifically, the compactness scores improved in general for California with great alternation in the layout of the two redistricting maps. California's 23rd District in the 112th Congress, the Ribbon of Shame, no longer exists in the current map. Together with the improved compactness scores and no changes in partisan bias, the outcomes of nonpartisan efforts to mitigate partisan gerrymandering in California remain unconvincing. What is certain is that, in spite of continued but slight Democratic bias, overall district compactness improved. The system in California still disadvantages Republicans, however, and Republicans

win about three fewer seats than is fair under expectations of no partisan bias. In addition, the average Republican district in California is less compact than the average Democratic district for population compactness. The increase in overall compactness scores, suggesting more regularly shaped CDs, might be attributable to the shift in the responsibility for enacting districting plans from the California State Legislature to an independent Citizens Redistricting Commission during this round of redistricting. The continued partisan disparity in compactness and bias, though, indicates that the Commission might not have been able to improve all of its objectives for California's electoral geography simultaneously. It is likely that small trade-offs were made between political bias and improved spatial compactness, resulting in the continued partisan bias but significant overall improvement in districts' overall spatial compactness.

In contrast, the new redistricting map for North Carolina is clearly not promising. A majority of districts in North Carolina decreased in their compactness scores, indicating less compact districts and more irregular boundaries in the new district map. In addition, the magnitude of partisan bias increased substantially, changing direction from favoring Democrats to favoring Republicans. This is disheartening, as North Carolina's checkered past with other types of gerrymandering, like racial gerrymandering, is well known (Thompson 2002). Overall, this shift results in more wasted Democratic votes in less compact districts. Taken together, these indicators provide a classic image of partisan gerrymandering that has been observed throughout U.S. political history and is clearly occurring in North Carolina.

In general, the comparison between the NMMI and Reock indexes for individual districts revealed that NMMI is more effective at indicating when districts' shapes belie a noncompact population distribution. We found that existence of natural barriers might result in relatively low Reock scores, even though gerrymandering might not be evident. On the other hand, the NMMI can also be very low, owing to extremely dispersed patterns of high-density population centers within the district, regardless of the geographical shape of district boundaries. In general, we suggest that one should use measures in the MI family to evaluate the compactness of districting plans, as they can accommodate more information. This additional information makes them useful packing situations where other compactness measures fail. Individual district-level comparisons demonstrate that MI

measures offer a new, more robust set of analytical tools for compactness analysis that can be easily customized to assess districting plans with or without supplemental population information.

Future efforts should consider applying these compactness measures to more districting plans, and expanding this type of study to nationwide analysis might provide more validation for these conclusions. In addition, it will be beneficial to incorporate political or demographic traits (e.g., race, party registration, ideology, income) into compactness analysis to gain a more comprehensive perspective of the social dimensions of district compactness. Additional spatial models sensitive to spatial variation in partisan bias, like local indicators of voter power or efficacy, might also improve the analysis. All of these political, social, and spatial factors might reflect particular patterns inherent in the redistricting process and can serve as major motivation of gerrymandering. As such, the NMMI provides a new tool to measure compactness while considering these properties and opens up new opportunities for analysis and insightful suggestions about the presence of unfair partisan gerrymandering.

References

- Angel, S., and J. Parent. 2011. Non-compactness as voter exchange: Towards a constitutional cure for gerrymandering. *Northwestern Interdisciplinary Law Review* 4:89–145.
- Arrow, K. J. 2012. *Social choice and individual values*. 3rd ed. New Haven, CT: Yale University Press.
- Baço, F., V. Lobo, and M. Painho. 2005. Applying genetic algorithms to zone design. *Soft Computing* 9 (5): 341–48.
- Batkins, S. B. 2008. Cracking, packing, and shacking: Resolving gerrymandering disputes at the ballot box? http://works.bepress.com/samuel_batkins/1 (last accessed 4 May 2015).
- Bozkaya, B., E. Erkut, and G. Laporte. 2003. A tabu search heuristic and adaptive memory procedure for political districting. *European Journal of Operational Research* 144 (1): 12–26.
- Brookes, R. H. 1960. The analysis of distorted representation in two-party single-member elections. *Political Science* 12:158–67.
- Brunell, T. L., and B. Grofman. 2005. Evaluating the impact of redistricting on district homogeneity, political competition, and political extremism in the U.S. House of Representatives, 1962–2002. Unpublished manuscript, University of California, Irvine, CA.
- Chambers, C., and A. Miller. 2010. A measure of bizarreness. *Quarterly Journal of Political Science* 5 (1): 27–44.
- Chen, J., and J. Rodden. 2013. Unintentional gerrymandering: Political geography and electoral

- bias in legislatures. *Quarterly Journal of Political Science* 8:239–69.
- Chou, C., S. Kimbrough, J. Sullivan-Fedock, C. J. Woodard, and F. H. Murphy. 2012. Using interactive evolutionary computation (IEC) with validated surrogate fitness functions for redistricting. Paper presented at the fourteenth International Conference on Genetic and Evolutionary Computation, Philadelphia, PA.
- Douzet, F., T. Kousser, and K. P. Miller. 2008. *The new political geography of California*. Berkeley, CA: Berkeley Public Policy Press.
- Engstrom, R. L. 1986. Repairing the crack in New Orleans' black vote: VRA's results test nullifies "gerrymander." *Publius: The Journal of Federalism* 16 (4): 109–22.
- Forest, B. 2013. Redistricting and the elusive ideals of representation. *Political Geography* 32:15–17.
- Fryer, R. G., Jr., and R. T. Holden. 2007. *Measuring the compactness of political districting plans*. Working Paper No. 13456, National Bureau of Economic Research, Cambridge, MA.
- Garfinkel, R. S., and G. L. Nemhauser. 1970. Optimal political districting by implicit enumeration techniques. *Management Science* 16 (8): B-495–B-508.
- Gelman, A., and G. King. 1994. A unified method of evaluating electoral systems and redistricting plans. *American Journal of Political Science* 38 (2): 514–54.
- Gelman, A., G. King, and A. Thomas. 2010. JudgeIt II: A program for evaluating electoral systems and redistricting plans. <http://gking.harvard.edu/judgeit> (last accessed 4 May 2015).
- Grofman, B. 1983. Measures of bias and proportionality in seats–votes relationships. *Political Methodology* 9 (3): 295–327.
- Grofman, B., and G. King. 2007. The future of partisan symmetry as a judicial test for partisan gerrymandering after *LULAC v. Perry*. *Election Law Journal* 6 (1): 2–35.
- Grofman, B., and P. Selb. 2009. A fully general index of political competition. *Electoral Studies* 28 (2): 291–96.
- Gudgin, G., and P. J. Taylor. 1976. *Seats, votes, and the spatial organisation of elections*. Colchester, UK: ECPR Press.
- Hayes, D., & S. C. McKee. 2009. The participatory effects of redistricting. *American Journal of Political Science* 53 (4): 1006–23.
- Hess, S. W., J. Weaver, H. Siegfeldt, J. Whelan, and P. Zitlau. 1965. Nonpartisan political redistricting by computer. *Operations Research* 13 (6): 998–1006.
- Hodge, J. K., E. Marshall, and G. Patterson. 2010. Gerrymandering and convexity. *The College Mathematics Journal* 41 (4): 312–24.
- Hofeller, T., and B. Grofman. 1990. Comparing the compactness of California congressional districts under three different plans: 1980, 1982, and 1984. In *Political gerrymandering and the courts*, ed. B. Grofman, 281–88. New York: Algora.
- Hojati, M. 1996. Optimal political districting. *Computers & Operations Research* 23 (12): 1147–61.
- Humphreys, M. 2011. Can compactness constrain the gerrymander? *Irish Political Studies* 26 (4): 513–20.
- Katz, R. 1980. *A theory of parties and electoral systems*. Baltimore: Johns Hopkins University Press.
- King, G., and R. X. Browning. 1987. Democratic representation and partisan bias in Congressional elections. *American Political Science Review* 81 (4): 1251–73.
- Kogan, V., and E. McGhee. 2012. Redistricting California: An evaluation of the Citizens Commission final plans. *California Journal of Politics & Policy* 4 (1): 1–22.
- Li, W., T. Chen, E. A. Wentz, and C. Fan. 2014. NMMI: A mass compactness measure for spatial pattern analysis of areal features. *Annals of Association of American Geographers* 104 (6): 1116–33.
- Li, W., M. F. Goodchild, and R. Church. 2013. An efficient measure of compactness for two-dimensional shapes and its application in regionalization problems. *International Journal of Geographical Information Science* 27 (6): 1227–50.
- Lublin, D. 1999. *The paradox of representation: Racial gerrymandering and minority interests in Congress*. Princeton, NJ: Princeton University Press.
- Manninen, D. L. 1973. *The role of compactness in the process of redistricting*. Seattle: University of Washington.
- McConnell, M. W. 2000. Redistricting cases: Original mistakes and current consequences. *Harvard Journal of Law & Public Policy* 24:103–17.
- McKee, S. C. 2008. Redistricting and familiarity with U.S. House candidates. *American Politics Research* 36 (6): 962–79.
- . 2013. Political conditions and the electoral effects of redistricting. *American Politics Research* 41 (4): 623–50.
- McKee, S. C., J. M. Teigen, and M. Turgeon. 2006. The partisan impact of congressional redistricting: The case of Texas, 2001–2003. *Social Science Quarterly* 87 (2): 308–17.
- Mehrotra, A., E. L. Johnson, and G. L. Nemhauser. 1998. An optimization based heuristic for political districting. *Management Science* 44 (8): 1100–14.
- Miller, P., and B. Grofman. 2013. Redistricting commissions in the western United States. *UC Irvine Law Review* 3:637–68.
- Niemi, R. G., B. Grofman, C. Carlucci, and T. Hofeller. 1990. Measuring compactness and the role of a compactness standard in a test for partisan and racial gerrymandering. *The Journal of Politics* 52 (4): 1155–81.
- Parker, F. R. 1985. Mississippi congressional redistricting case: A case study in minority vote dilution. *Howard Law Journal* 28:397–415.
- Polsby, D. D., and R. D. Popper. 1991. Third criterion: Compactness as a procedural safeguard against partisan gerrymandering. *Yale Law & Policy Review* 9:301–53.
- Quinton, N. 2013. Political criteria, political praxis. *Political Geography* 32:18–20.
- Reock, E. C. 1961. A note: Measuring compactness as a requirement of legislative apportionment. *Mid-west Journal of Political Science* 5:70–74.
- Sauer, C. O. 1918. Geography and the gerrymander. *The American Political Science Review* 12 (3): 403–26.
- Silverberg, K. 1995. Illegitimacy of the incumbent gerrymander. *Texas Law Review* 74:913–41.

- Taagepera, R. 1973. Seats and votes: A generalization of the cube law of elections. *Social Science Research* 2:257–75.
- Taylor, P. J., and R. J. Johnston. 1979. *Geography of elections*. New York: Penguin Books.
- Thompson, D. T. 2002. *Congressional redistricting in North Carolina: Reconsidering traditional criteria*. El Paso, TX: LFB Scholarly Publishing.
- Tufte, E. 1973. The relationship between seats and votes in two-party systems. *American Political Science Review* 67 (2): 540–54.
- Weaver, J. B., and S. W. Hess. 1963. A procedure for non-partisan districting: Development of computer techniques. *The Yale Law Journal* 73 (2): 288–308.
- Webster, G. R. 2013a. Normative goals and demographic realities. *Political Geography* 32:21–22.
- . 2013b. Reflections on current criteria to evaluate redistricting plans. *Political Geography* 32:3–14.
- Young, H. P. 1988. Measuring the compactness of legislative districts. *Legislative Studies Quarterly* 13 (1): 105–15.

Correspondence: GeoDa Center for Geospatial Analysis and Computation, School of Geographical Sciences and Urban Planning, Arizona State University, Tempe, AZ 85287–5302, e-mail: cfan13@asu.edu (Fan); wenwen@asu.edu (Li); levi.john.wolf@asu.edu (Wolf); soe.myint@asu.edu (Myint).